

Lists, Tuples & Dictionaries

Pre-Assessment 3 · 13 questions · 45 points

Name: _____

Date: _____

Score: / 45

This is a **pre-assessment**. It shows what you already know **before** we start this unit, so it is fine if some questions cover things you have not learned yet. Try every question you can. If you have no idea at all, write **“not yet”** and move on.

Work on your own, without a computer. You have about **45 minutes**. When a question asks what a program prints, write exactly what Python would print, one line of output per answer line. If the code would crash, write the name of the error (for example, `TypeError`). Show your thinking; partial credit is given.

1. Snack list

4 points

`snacks = ["chips", "apple", "cookie", "grapes", "pretzel"]`. Write what each expression gives.

Expression	Result
<code>snacks[0]</code>	
<code>snacks[-1]</code>	
<code>snacks[1:3]</code>	
<code>len(snacks)</code>	

2. Change it

2 points

```
letters = ["c", "a", "t"]
letters[0] = "b"
print(letters)
word = "cat"
word[0] = "b"
```

(a) What does line 3 print?

(1 pt)

(b) What happens on line 5, and why is it different from line 2?

(1 pt)

3. Playlist Editor

3 points

Write exactly what this program prints.

```
playlist = ["Song A", "Song B", "Song C"]
playlist.append("Song D")
playlist.insert(0, "Intro")
playlist.remove("Song B")
last = playlist.pop()
print(playlist)
print(last)
```

4. Sort it out

3 points

Write exactly what this program prints.

```
nums = [5, 2, 9, 1]
nums.sort()
print(nums)
nums.reverse()
print(nums[0])
print(7 in nums, 2 not in nums)
```

5. Temperature report

4 points

Fill in the three blanks so the program finds the **total** and the **biggest** number without using `sum()` or `max()`. Then write what it prints.

```
temps = [18, 25, 21, 30, 16]
total = 0
biggest = temps[0]
for t in _____:
    total = _____
    if _____:
        biggest = t
print(total, biggest)
```

It prints: _____

6. Points on a map

3 points

```
point = (3, 7)
x, y = point
print(x + y)
```

(a) What does this print?

(1 pt)

(b) What happens if we then run `point[0] = 5`?

(1 pt)

(c) Why might a programmer store a map point as a tuple instead of a list?

(1 pt)

7. Sentence Sorter

3 points

Write exactly what this program prints.

```
sentence = "my cat loves sunny windows"
words = sentence.split()
words.sort()
print(words)
print(" ".join(words))
longest = ""
for w in words:
    if len(w) > len(longest):
        longest = w
print(longest)
```

8. Pet Profile

4 points

```
pet = {"name": "Mochi", "type": "cat", "age": 3}
pet["age"] = 4
pet["color"] = "gray"
del pet["type"]
print(pet)
print(pet["name"])
print("type" in pet)
```

(a) Write exactly what the program prints.

(3 pts)

(b) What would happen if we added `print(pet["owner"])` at the end?

(1 pt)

9. Fruit Stand

4 points

Write exactly what this program prints.

```
stock = {"apples": 5, "pears": 0, "plums": 3}
for fruit, count in stock.items():
    if count > 0:
        print(fruit, count)
print(stock.get("kiwis", 0))
total = 0
for count in stock.values():
    total = total + count
print(total)
```

10. Word Counter

4 points

```
text = "red blue red green red blue"
counts = {}
for word in text.split():
    if word in counts:
        counts[word] = _____
    else:
        counts[word] = _____
print(counts)
```

(a) Fill in the two blanks.

(2 pts)

(b) Write what the program prints.

(2 pts)

11. List or dictionary?

2 points

For each kind of data, write **list** or **dictionary** and give a short reason.

(a) The order in which runners finished a race.

(1 pt)

(b) Each student's locker number.

(1 pt)

12. Class Directory

4 points

```
students = [
    {"name": "Ana", "grade": 5, "club": "chess"},
    {"name": "Ben", "grade": 6, "club": "art"},
    {"name": "Cy", "grade": 5, "club": "art"},
]
for s in students:
    if s["club"] == "art":
        print(s["name"], s["grade"])
print(students[0]["club"])
```

(a) Write exactly what the program prints.

(3 pts)

(b) Write one line of code that adds a new student: Dev, grade 6, chess club.

(1 pt)

13. Trivia Quiz Show

5 points

Use this list of dictionaries:

```
quiz = [  
    {"question": "What is 3 x 4?", "answer": "12"},  
    {"question": "Mix blue and yellow to get?", "answer": "green"},  
    {"question": "How many legs does a spider have?", "answer": "8"},  
]
```

Write code that asks each question, accepts messy answers (so " Green " counts as correct), keeps score, and at the end prints You got 2 out of 3! (with the real numbers).

END OF ASSESSMENT